

LAB 3: COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

CARBON TRACKER — HOUSEHOLD CARBON FOOTPRINT & TARGET TRACKER

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1. Architecture Selection

We chose Layered Architecture for the Carbon Tracker system.

2. Architectural Choice

The Carbon Tracker system is organized into Presentation, Business/Application and Data layers. The Presentation Layer handles resident and sustainability-coach interactions. The Business/Application Layer performs carbon calculation, target and progress tracking, and coach review. The Data Layer stores household consumption and historical information.

3. Two Scenario-Related Reasons

1. Clear separation of concerns: user interaction is separated from carbon calculation, target/progress logic and data storage, making each responsibility easier to understand and maintain.
2. Easier maintenance and testing: changes to emission-calculation rules or target logic can be made in the application layer without changing the user interface or repository implementation.

4. Security Advantage

The UI does not need direct access to household consumption records. Requests can pass through application services where validation and authorization can be applied before sensitive household data is stored or retrieved.

5. Performance Benefit

Carbon calculations and progress requests can be handled in the application layer while the repository performs focused data operations. This avoids unnecessary direct database access from the user interface.

6. Component Identification

- Resident / Coach UI
- Carbon Calculation Service
- Target & Progress Service
- Coach Review Service
- Consumption & History Repository

7. Interface Identification

- Calculation API
- Target API
- Review API
- Data Access
- History Access
- Consumption Data
- History Data

8. Component Interactions / Data Flow

Resident / Coach UI communicates with the Carbon Calculation Service, Target & Progress Service and Coach Review Service. These application components use the Consumption & History Repository for consumption, progress and review/history data. The diagram uses solid connectors for internal service communication and labels the interactions with API calls, database queries and data-flow descriptions.

9. UML Component Diagram

Carbon Tracker — UML Component Diagram

Layered Architecture

